Selective Entry and Structured Electricity-Recovery Performance in Japan's Waste-Incineration Fleet: A Facility-Level Panel Study

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Abstract

Japan relies heavily on municipal waste incineration, yet by fiscal year (FY) 2024 only 41.1% of facilities in the panel are flagged as power-generating. This creates a two-margin modernization problem. Facilities outside electricity recovery face an entry margin, while existing generators face a conditional-performance margin. Fleet-average studies blur those margins, and generator-only studies miss the entry margin altogether. Using Ministry of the Environment data for FY2005–FY2024, this paper estimates both margins in one national facility panel. It asks whether first observed entry into generation is associated with prior-year age and capacity, how electricity recovered per tonne is associated with observed facility and operating conditions among identifiable generators, and whether those two margins support one average-fleet modernization interpretation. Entry is selective rather than diffuse: facilities older than ten years are less likely to record transition in the next fiscal year, while larger facilities are more likely to do so. Among operating generators, electricity recovery intensity is lower at older plants and higher at larger, more fully utilized ones, while between-facility heterogeneity dominates within-facility movement. An aggregate view therefore understates both the selectivity of entry and the persistence of performance hierarchy. These patterns are descriptive within the linked samples, not causal estimates of a single modernization mechanism. For municipal fleet planning, non-generators and mature generators should not be managed as one average segment.

Keywords: waste incineration; waste-to-energy; Japan; energy recovery; facility panel; transition

1 Introduction

Japan operates one of the world's most incineration-dependent municipal waste systems, yet many facilities still burn waste without generating electricity from the heat they produce (Ministry of the Environment Japan, 2022; Uno, 2015; Tabata & Tsai, 2016; Sakai et al., 2011). In fiscal year (FY) 2024, 41.1% of facilities in the panel are flagged as power-generating, leaving most facilities outside electricity recovery. This is not a marginal technical detail. For decades, Japanese municipal waste governance has relied on thermal treatment for hygienic disposal, volume reduction, and local waste autonomy, while limited landfill space and strict environmental controls pushed municipalities toward sophisticated incineration infrastructure (Brunner & Rechberger, 2015; Sakai et al., 2011). Sectoral planning now treats the remaining electricity-generation gap as part of the waste sector's decarbonization challenge (Yamada et al., 2023; Greenhouse Gas Inventory Office of Japan & Ministry of the Environment Japan, 2024). The transition problem is therefore not whether Japan uses incineration. It is whether observed entry into electricity recovery is patterned across facilities and how electricity recovery intensity varies once generation is present.

Aggregate fleet summaries flatten two distinct questions. One is extensive: which facilities record observed transition into power generation at all? The other is intensive: among plants that already generate, which facilities achieve high electricity recovery per tonne? The system can therefore look uniformly slow in the mean while combining selective entry at one end of the fleet with persistent performance hierarchy at the other.

Research questions

This paper asks two descriptive empirical questions and one interpretive synthesis question. In the list, RQ denotes research question:

RQ1: Entry margin. Among coded facilities first observed without power generation, which prior-year age and capacity profiles are associated with first observed reporting of power generation in the following fiscal year?

RQ2: Generator-performance margin. Among identifiable operating generators, how is electricity recovered per tonne associated with facility age, design capacity, utilization, heating value, and common fiscal-year conditions?

RQ3: Interpretive synthesis. Taken together, do the observed entry pattern and generator-frame performance associations support one average-fleet modernization interpretation, or do they indicate distinct adoption and performance problems within the same incineration system?

The contribution is not a causal estimate of retrofit effects or a technical optimization model. It is a linked facility-panel decomposition that shows why observed entry into generation and conditional generator performance should be modeled separately before being interpreted together. The paper uses two linked samples: one follows non-generating facilities until they first report electricity generation, and the other compares identifiable operating generators over time. In the first sample, observed transition is selective toward younger and larger facilities. In the second, electricity recovery intensity is strongly structured by age, scale, and utilization, while within-facility movement remains limited relative to between-facility heterogeneity. The same fleet can therefore look merely slow in aggregate while containing two different problems: selective entry into generation and persistent generator hierarchy. For municipal planning, that distinction changes the first diagnostic step. Facilities outside electricity recovery and mature generators should be evaluated as different asset-management questions before they are summarized as one fleet.

A natural expectation is that younger and larger plants will have advantages. The contribution is not that age and scale matter in isolation. It is that the same administrative fleet shows those advantages on two different margins: younger and larger facilities are overrepresented in observed entry into generation, while operating generators remain stratified after entry. That two-margin structure is what an aggregate fleet mean or generator-only study would miss.

The gap addressed here is narrower than a generic claim that Japan has been understudied. Waste-to-energy research can describe fleet trajectories, and generator-only studies can explain conditional performance once plants already operate as generators. What remains uncommon is one linked

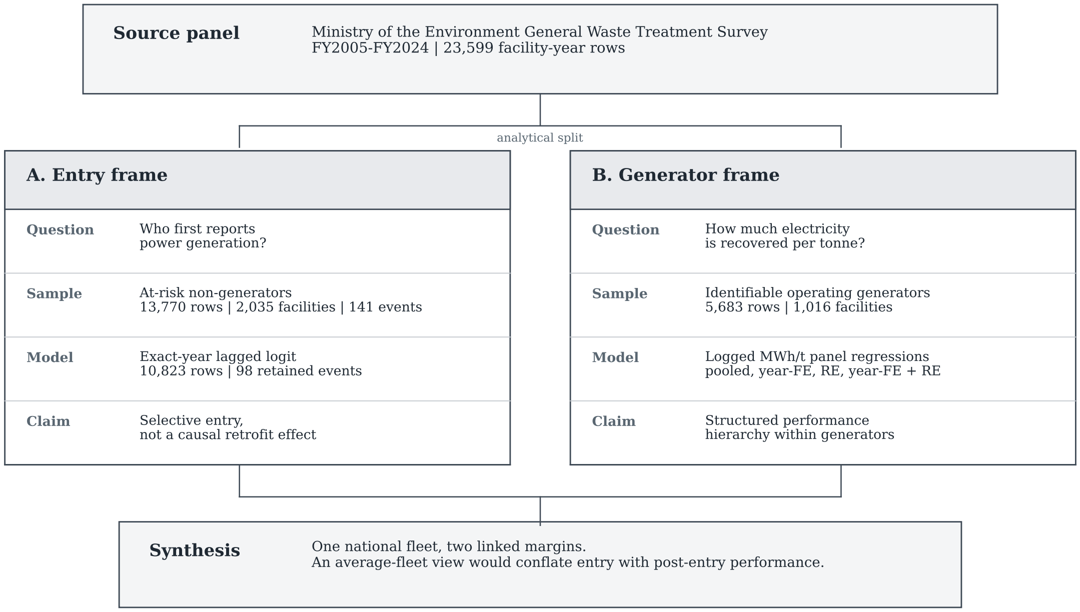

Figure 1: Analytical design separating the source panel into entry and generator-performance margins before interpreting both margins together.

municipal-fleet analysis that estimates both margins and asks whether they point to the same modernization bottleneck. Japan makes that contrast visible because persistent non-generators, old small plants, and a more modern generating segment coexist in the same administrative system. In such mixed fleets, adoption and conditional performance should be modeled separately before they are interpreted together.

The rest of the paper proceeds as follows. Section 2 positions the paper against the literature most relevant to the analytical split. Section 3 introduces the two linked analytical frames and the main estimation choices. Section 4 reports the adoption and electricity-recovery results in sequence, then ties them together. Section 5 interprets the combined finding, explains what the data still cannot identify, and states a short set of evidence-consistent implications. Section 6 concludes.

2 Literature Positioning

This paper speaks to three literatures that often meet in practice but start from different empirical positions: waste-to-energy systems, facility-level performance analysis, and infrastructure lock-in. The review is organized around the paper's central split between entry into generation and conditional performance after entry. The relevant gap is not simply that Japan needs another case study. It is that existing studies often describe fleets in aggregate or explain generator performance conditional on operation, yet rarely estimate both margins in one linked municipal-fleet design.

Work on waste-to-energy systems often documents national trajectories, technology choices, or lifecycle implications of thermal treatment (Astrup et al., 2009; Astrup et al., 2015; Brunner & Rechberger, 2015; Sun et al., 2018). Japan-specific work has mostly emphasized technology upgrading, heat-use constraints, and sectoral decarbonization scenarios rather than facility-level transition modeling (Uno, 2015; Tabata & Tsai, 2016; Yamada et al., 2023). Policy-facing work reaches a similar conclusion from a different angle: waste-to-energy is treated as useful only when embedded in a wider waste hierarchy and resource-recovery strategy rather than as a stand-alone justification for thermal treatment (European Commission, 2017; Sakai et al., 2011). This literature is indispensable for understanding why energy recovery matters, but it usually treats the incineration fleet as a sectoral category. It can show whether energy recovery is expanding, whether heat supply remains constrained, or whether the sector matters for net-zero planning. It is less well suited to distinguishing facilities that enter generation from those that do not.

Facility-level performance studies are closer to the present design, but they typically begin after entry has already occurred. Studies in Taiwan, for example, evaluate operating incinerators by decomposing waste-treatment, electricity-generation, or revenue performance within existing plants (Chen et al., 2012; Yeh, 2020). Other work focuses on energy-recovery criteria, plant scale, heat use, and the system consequences of different waste-to-energy configurations (Grosso et al., 2010; Münster & Meibom, 2010). Recent Chinese plant-level work is also highly informative about performance differentials inside the generating segment and the effectiveness of upgrading strategies at scale (Cui et al., 2026; Liu et al., 2025; Han et al., 2025). Those studies help define the intensive-margin question. They do not directly answer which non-generating facilities enter generation in the first place.

The present paper therefore uses those comparator papers as a method-positioning map rather than as templates to copy. The map below makes the adaptation logic explicit.

Comparator adaptation of research logic

Source family	Borrowed logic	Japan-specific adaptation	Not claimed here
Cui et al. (2026)	Incinerators form a performance hierarchy, not interchangeable plants	The Japan panel is read through facility heterogeneity and persistent hierarchy	No optimization frontier or ranking of Japanese plants on the same technical frontier
Liu et al. (2025)	Waste-energy systems should be judged by effectiveness, not expansion alone	Effectiveness is split into entry into generation and performance after entry	No China-style waste-energy-carbon development model
Han et al. (2025)	Resource recovery and sustainability depend on upgrade pathways, not incineration alone	Resource-recovery framing is kept while using variables available in the Japanese administrative panel	No pollutant-control technology model because comparable plant-level variables are unavailable
Chen et al. (2012) and Yeh (2020)	Facility-level incinerator performance can be compared after operation begins	The generator frame models electricity recovered per tonne across identifiable plants	No data envelopment analysis frontier or electricity-revenue inefficiency model
Grosso et al. (2010) and Münster and Meibom (2010)	Energy recovery should be interpreted in a wider energy-system context	MWh/t and a grid-emission-factor control provide applied performance context	No full R1 efficiency calculation or energy-system optimization model

This matters for interpretation because the paper is not trying to be a weaker version of any one comparator. Its foundation is a deliberate combination: facility heterogeneity from the high-profile incineration literature, transition timing from event-history methods, and conditional performance comparison from panel analysis. The originality claim is modest but clear: applying that combination to Japan shows why non-generators and operating generators should be read as two linked but different modernization margins.

The lock-in literature adds a different expectation. Infrastructure performance may be shaped by durable design choices, inherited scale, and institutional arrangements rather than by frequent late-life reversals at mature facilities (Unruh, 2000; Geels, 2004; Seto et al., 2016). The useful implication here is not that incineration plants are literally irreversible. It is that cross-facility differences may matter more than repeated performance resets. In municipal infrastructure, that persistence can also be institutional rather than purely technical: plant networks are shaped by jurisdictional boundaries, merger histories, charging regimes, and the political difficulty of reorganizing service territories (Rausch, 2006; Sakai et al., 2008; Sakai et al., 2011). That expectation is only partially visible if entry into generation and performance within generation are never separated.

Read together, the literatures imply a practical identification problem for fleet studies. Fleet-level work can show aggregate progress but cannot tell whether low average performance reflects many non-generators, weak generator performance, or both. Generator-only work can estimate performance correlates after entry, but not who stays outside generation. Lock-in work explains why mature infrastructure may remain stratified, but not whether the key margin lies before or after entry. The contribution of this paper is to use one linked panel to expose those blind spots without treating them as one average process.

3 Data and Design

The analysis uses the Ministry of the Environment's General Waste Treatment Survey for FY2005–FY2024 (Ministry of the Environment Japan, 2022). From 23,599 facility-year rows, the coded frame retains 19,827 observations across 2,948 identifiable facilities. The paper then separates two linked samples because one sample cannot answer both parts of the transition problem. Those frames are not only data filters. They are the analytic structure that keeps observed entry into generation separate from conditional performance after entry. The survey is useful for this purpose because it covers both generating and non-generating facilities inside the same administrative system. That makes it possible to ask a question that many sector studies cannot ask cleanly: which facilities record observed transition into generation, and how do facilities perform once they are already inside the generating segment? The two samples differ because the questions differ: non-generators reveal who enters electricity recovery, while generators reveal performance differences after entry.

The design intentionally combines established empirical building blocks rather than inventing a new estimator. The adoption layer follows discrete-time event-history logic for a binary transition observed in annual administrative data (Allison, 1982; Beck et al., 1998). The generator layer follows facility-level performance studies that compare operating incinerators after entry, while using panel regressions to summarize persistent cross-facility structure (Chen et al., 2012; Yeh, 2020; Wooldridge, 2010). What is different here is the way those pieces are linked: the paper first models observed entry into generation, then separately models electricity recovered per tonne among identifiable generators, and only then asks whether a single average-fleet interpretation is adequate. This makes the framework similar in spirit to plant-level optimization and performance papers, but distinct in its Japan-specific two-margin diagnostic focus.

Research-question-to-model bridge

Question	Main frame and model	What it can show	What it cannot show
RQ1: entry	Coded non-generators; exact one-fiscal-year lagged logit hazard with year fixed effects	Which prior-year age and capacity profiles are associated with first reporting generation	A causal retrofit effect, full capital-history mechanism, or unrestricted fleet-wide modernization
RQ2: generator performance	Identifiable generators; logged MWh/t panel regressions using pooled, year-FE, RE, and year-FE + RE specifications	Whether electricity recovered per tonne is structured by age, scale, utilization, heating value, and common year conditions	Full thermodynamic efficiency or the causal effect of changing age, scale, utilization, or technology
RQ3: interpretive synthesis	Two linked but non-identical frames read together	Whether one average-fleet interpretation hides two different bottlenecks	A strict causal chain from adoption into generation to later generator performance

The first frame is the coded adoption frame. It includes facilities first observed without power generation and follows them until they either record observed transition into generation or remain non-generating in the panel window. After excluding left-censored facilities already generating in their first observed year, the adoption risk set contains 13,770 facility-years across 2,035 facilities, with 141 observed first-adoption events.

That 141-event count is the descriptive adoption universe used for event-rate summaries and the pathway audit. The main hazard model below uses a stricter exact one-fiscal-year lagged subset with 98 retained events; the 42 timing-ambiguous non-adjacent events and 1 unresolved event remain in pathway and sensitivity evidence rather than being forced into the headline hazard.

In practical terms, the adoption model asks whether a facility that was still non-generating in one year first reports generation in the following fiscal year. Technically, it is an exact one-fiscal-year lagged discrete-time logit hazard estimated on 10,823 observations across 1,911 facilities and 98 retained events. Predictors are prior-year age band and prior-year design capacity, with year fixed effects and facility-clustered standard errors. A more saturated year-plus-prefecture fixed-effects model is retained as sensitivity evidence rather than used as the main specification because it would estimate 64 parameters with 98 retained events, or 1.53 events per parameter. The primary year fixed-effects model estimates 18 parameters, or 5.44 events per parameter. The exact-year restriction is important because official facility codes are missing in FY2010–FY2012; broader previous-observed-coded-row estimates are also reported only as sensitivity evidence. This is an observed-transition model, not a complete structural model of all possible modernization pathways. The design follows grouped event-history logic: each coded facility-year contributes to the risk set until first event occurrence (Allison, 1982; Beck et al., 1998). That distinction matters because the paper is not estimating a continuous engineering retrofit process. It estimates the probability that a facility first records entry into power generation in the next fiscal year, conditional on still being at risk. The lagged predictor structure ensures that age band and capacity are measured before the observed event rather than on the event row itself.

Formally, let $A_{it} = 1$ if facility i first reports power generation in fiscal year t, conditional on still being in the at-risk set $R_{it} = 1$. The main adoption model is:

$$\begin{aligned}\Pr(A_{it} = 1 \mid R_{it} = 1) = \operatorname{logit}^{-1}\big[&\alpha + \beta_1 I(\mathrm{Age}_{i,t-1} = 10\text{–}20) + \beta_2 I(\mathrm{Age}_{i,t-1} = 20\text{–}30) \\ &+ \beta_3 I(\mathrm{Age}_{i,t-1} \geq 30) + \beta_4 \mathrm{Capacity100}_{i,t-1} + \gamma_t\big].\end{aligned} \tag{1}$$

Here, i indexes facilities, t indexes fiscal years, and $I(\cdot)$ is an indicator equal to 1 when the stated condition is true. The omitted age group is 0–10 years. γ_t absorbs common fiscal-year differences, and Capacity100 measures design capacity in 100 tonnes per day (t/day) units. Table 2 reports average marginal effects (AMEs) in percentage points (pp), not log-odds coefficients. A value of -1.67 pp therefore means that the annual probability of first reporting generation is 1.67 percentage points lower than the 0–10 year reference group, conditional on the model covariates. The prefecture fixed-effects variant is reported in the supplement as a sensitivity check rather than as the primary estimate.

Age and capacity are therefore treated as observable pre-event facility profiles, not isolated causal mechanisms. They may also proxy for unobserved municipal finance, procurement timing, consolidation, routing arrangements, technology vintage, or renewal planning. The adoption model is useful because those profiles describe which facilities are most likely to appear in observed entry; it cannot determine which institutional or engineering channel produced that association.

The second frame is the canonical generator frame. It contains operating facilities with positive throughput and positive power output, after standard cleaning and electricity-recovery bounding. The operating-generation sample contains 6,660 rows before identifier and regression cleaning. Of those, 907 rows lack official facility codes and are excluded from the canonical regression frame, leaving 5,683 observations across 1,016 facilities. The dependent variable is log electricity recovered per tonne processed. Main predictors are facility age, design capacity, capacity utilization, waste heating value, and a grid-emission factor control. In simpler terms, the outcome is electricity recovered per tonne of waste. The raw ratio is megawatt-hours (MWh) generated divided by tonnes processed, clipped to 0.01–0.80 megawatt-hours per tonne (MWh/t), and then log-transformed so that a few unusual administrative records do not dominate the comparison. Capacity utilization is also capped at 1.0. The paper uses this measure as electricity recovery intensity, not as a full thermodynamic efficiency measure.

Let q_{it} be the clipped electricity recovery ratio and let $y_{it} = \log(q_{it})$. The compact model is:

$$y_{it} = \alpha + X_{it}'\beta + \gamma_t + u_i + \varepsilon_{it}, \tag{2}$$

where X_{it} contains facility age, design capacity in 100 t/day units, capped capacity utilization, heating value, and the grid-emission factor (EF). The EF is measured as kilograms of carbon dioxide per kilowatt-hour (kg-CO_2/kWh). The four reported models are nested descriptive versions of this equation. Model 1 omits both γ_t and u_i and compares facilities overall with pooled ordinary least squares (OLS). Model 2 includes fiscal-year fixed effects (FE) γ_t, which absorb year-specific shocks common to the fleet. Model 3 includes a facility-specific random intercept u_i, reported as a random-effects (RE) panel specification, which summarizes persistent facility-level differences. Model 4 includes both year FE and the random intercept. All four models use facility-clustered standard errors to avoid treating repeated observations from the same facility as fully independent (Wooldridge, 2010).

The random-effects specifications are not used because they solve all unobserved-facility concerns. They are used because the paper's intensive-margin question is descriptive cross-facility structure, while a pure facility fixed-effects interpretation would rely on limited within-facility movement and would identify a different question.

Because the dependent variable is logged, small coefficients can be read approximately as percentage changes in electricity recovered per tonne. For example, the pooled age coefficient of -0.0279 implies about 2.8% lower electricity recovered per tonne for each additional facility year, conditional on the listed variables. The pooled capacity coefficient of 0.0874 implies about 9.1% higher electricity recovered per tonne for each additional 100 t/day of design capacity. The random-effects models are used descriptively; they do not prove that unobserved facility traits are unrelated to age, capacity, technology, or municipal context.

The electricity-recovery results should therefore be read as conditional patterns within the canonical regression frame, not as estimates for the entire generating fleet. This frame is intentionally narrower than the operating-generator universe because the argument depends on facility-level comparison over time. Rows without official facility codes cannot support that comparison. The frame is better understood as the canonical identifiable generator sample than as a census of all generation activity. The supplement compares included and uncoded operating-generator rows; the uncoded rows are concentrated in FY2010–FY2012, so period comparisons are treated as coded-window diagnostics rather than as a clean Fukushima-era identification design. The paper also uses a bounded electricity-recovery metric because the empirical question is not boiler thermodynamics in isolation, but administrative performance in electricity recovered per tonne processed. That puts the paper closer to applied energy-recovery studies than to plant-level engineering optimization alone (Grosso et al., 2010; Münster & Meibom, 2010).

Two administrative-data checks are reported in the supplement. First, a small set of official facility codes appears more than once within the same fiscal year, so a composite identifier (ID) sensitivity appends facility names to affected duplicate codes. Second, heating value is treated as a noisy control and is checked under plausible-value restrictions. Neither sensitivity changes the main sign pattern or the substantive interpretation. These checks are reported as disclosure and robustness evidence, not as an attempt to remove all administrative uncertainty from the source panel.

The two layers belong in one paper because they answer sequential parts of the same modernization problem. The adoption layer identifies which facilities appear to enter the generating regime. The electricity-recovery layer identifies whether large performance gaps remain once facilities are already inside that regime. Without the first layer, the paper would reduce transition to generator performance alone. Without the second, it would say who enters generation but not whether major electricity-recovery differences remain inside the generating segment.

The design is diagnostic rather than structural. It models observed transition within the coded risk set, not unrestricted fleet-wide modernization, and it models conditional generator performance within the canonical identifiable generator frame, not a causal effect of changing age, scale, utilization, or technology. The low within-facility variance ratio is used only to explain why cross-facility descriptive comparison remains informative; it does not resolve all fixed-effects concerns or turn the random-effects models into causal estimates. The defended claim is therefore narrower: under explicit sample limits, the two frames reveal different bottlenecks in the same fleet.

4 Results

4.1 Adoption into generation is selective rather than diffuse

The adoption results show a strongly selective transition pattern. In the risk set, annual event rates collapse after age 10 and rise sharply across capacity quartiles. Facilities aged 0–10 years account for 102 first-adoption events, while the three older age bands together account for only 39. By

Table 1: Linked analytical framework

Margin	Linked sample	Empirical question	Paper role
Adoption margin	Coded at-risk frame: 13,770 facility-years, 2,035 facilities, 141 observed first-adoption events	Which facilities record observed transition into generation?	Shows whether entry into generation is selective rather than diffuse
Electricity-recovery margin	Canonical generator frame: 5,683 observations across 1,016 operating generators	How does electricity recovered per tonne vary once generation already exists?	Shows whether mature generator performance remains structured
Synthesis	Two linked but non-identical analytical frames	Would one average-fleet view misstate the modernization bottleneck?	Shows why entry and mature performance should not be read as one average process

Note: the adoption margin is estimated with a lagged discrete-time hazard. The electricity-recovery margin is estimated with descriptive pooled, year fixed-effects (year-FE), and random-effects (RE) panel specifications.

capacity, the largest quartile accounts for 99 first-adoption events, whereas the smallest quartile records only 1. First adoption is also clustered in time: 109 of 141 observed first adoptions occur in FY2013–FY2019. That clustering should be read as an observed administrative transition pattern rather than as a separately identified policy shock or reporting change; the main hazard includes year fixed effects.

The descriptive event-rate and pathway statements use all 141 observed first-adoption events in the coded adoption frame. Table 2 reports the stricter exact-year hazard model, which retains 98 events after requiring an exact one-fiscal-year lag and complete model covariates.

The discrete-time logit hazard summarizes the same pattern in average marginal effects. The percentages below describe changes in the annual probability of first reporting generation, not changes in engineering efficiency. Relative to 0–10 year facilities, plants aged 10–20 years are about 1.67 percentage points less likely to record transition in the next fiscal year. Plants aged 20–30 years are about 1.94 percentage points less likely, and plants aged 30 years or more are about 1.24 percentage points less likely. Each additional 100 t/day of prior-year design capacity raises annual transition probability by about 0.45 percentage points. The sign pattern is stable in the alternative event-model checks reported in the supplement, including the saturated year-plus-prefecture fixed-effects model and the broader previous-observed-coded-row sensitivity.

These effects should be read within the coded at-risk frame, not as a model of all modernization activity in the Japanese fleet. Even within that narrower frame, the evidence does not show broad late-life conversion among old, small plants. Transition is concentrated among facilities that were already younger and larger before the event year. Older plants do transition in some cases, but they do so less often and do not define the main event pattern. The extensive margin therefore looks like selective observed entry rather than broad catch-up across the whole fleet. This matters because a descriptive fleet mean could be read as gradual modernization delayed by inertia, when the event pattern is more selective than gradual.

The pathway audit supports that interpretation without overstating mechanism. Among the 141 ob-

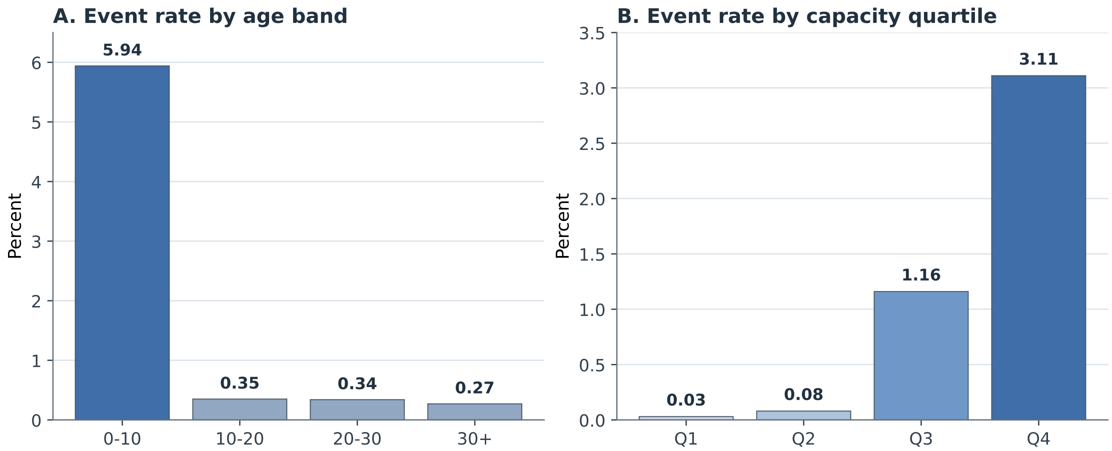

Figure 2: Observed adoption event rates by age band and capacity quartile in the coded at-risk frame.

Table 2: Main lagged hazard results for observed transition into generation

Variable	Average marginal effect (pp)	Standard error (pp)
Prior-year age 10–20 years (versus 0–10)	-1.67	0.25
Prior-year age 20–30 years (versus 0–10)	-1.94	0.39
Prior-year age 30+ years (versus 0–10)	-1.24	0.38
Prior-year capacity (per 100 t/day)	0.45	0.15
Model summary		
Observations		10,823
Facilities		1,911
First-adoption events		98
Pseudo-R-squared		0.1829

Note: entries are average marginal effects in percentage points (pp) from the main exact one-fiscal-year lagged logit hazard with year fixed effects and facility-clustered standard errors. The saturated year-plus-prefecture fixed-effects model is retained as sensitivity evidence because it has higher sparse-event pressure. The 98 events in this table are the exact-year hazard subset; the full descriptive adoption audit contains 141 observed first-adoption events.

served adoption events, 50 are classified as reset- or rebuild-like adjacent-year events, 36 as adjacent-year continuity-type upgrades, 12 as forward-dated or placeholder entries, 42 as timing-ambiguous non-adjacent coded-row events, and 1 as unresolved. This is descriptive pathway evidence, not mechanism identification. The adjacent-year event mix is more consistent with capital-intensive pathways than with diffuse late-life catch-up, but it does not uniquely identify replacement, major refurbishment, or new build as the single pathway. In the main text, the audit therefore functions as a credibility guard rather than as a coequal source of originality.

That distinction keeps the adoption result from becoming a story of technological inevitability. The event pattern does not imply that older plants never upgrade, nor that small facilities have no role in local waste management. It implies something narrower: within the coded at-risk frame, recorded entry into generation is not centered in the segment most likely to be described as lagging in simple fleet summaries. The modernization margin visible in the data is selective from the start.

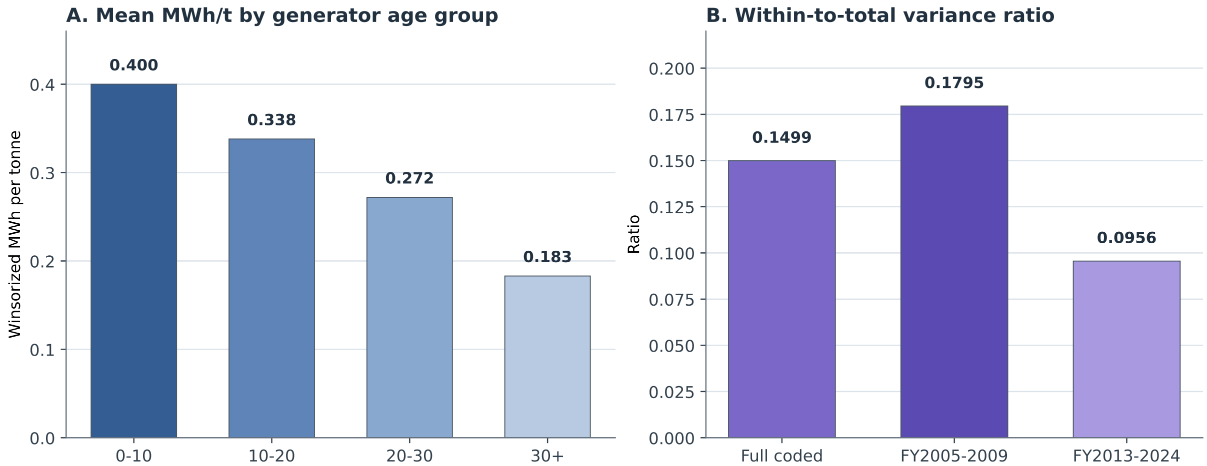

Figure 3: Mean electricity recovery intensity declines across generator age groups, while the within-to-total variance ratio stays low in the full coded sample and early/later coded windows.

4.2 Electricity recovery within generation is strongly structured

Within the canonical regression frame, the main message is that generators mostly differ from one another rather than repeatedly changing position over time. The within-to-total variance ratio of pooled log-efficiency is 0.1499, meaning that most variation is between facilities rather than within facilities over time. The ratio remains low in both the early coded window and later coded window, falling from 0.1795 in FY2005–FY2009 to 0.0956 in FY2013–FY2024. This is not a Fukushima identification design because FY2010–FY2012 operating-generator rows lack official facility codes. It shows more narrowly that the identifiable generator panel contains limited evidence of frequent large late-life movements that reshape the fleet distribution. It also does not isolate vintage effects from all other durable plant characteristics. More narrowly, it supports cross-facility descriptive comparison rather than clean causal isolation of vintage itself.

The coefficient patterns point in the same direction. Electricity recovery intensity is consistently lower at older facilities and higher at larger and more fully utilized ones. The magnitudes differ across models, but the sign pattern is stable. The emphasis is therefore on structured conditional association rather than on a causal estimate of what would happen if a plant's age, capacity, or utilization were changed by policy. This is consistent with earlier facility-level work showing that energy recovery performance is uneven across operating incinerators and that plant scale and operational intensity matter for output performance (Chen et al., 2012; Yeh, 2020; Grosso et al., 2010). What this paper adds is the linked comparison to the non-generating segment: the same fleet that shows selective entry at one margin also shows a stable hierarchy among mature generators at the other.

The electricity-recovery margin therefore looks structured rather than static. Facilities vary through utilization and operating conditions, but age and scale hierarchies remain strong in the observed data. This is where the paper diverges from a simple engineering-upgrade narrative. Recent large-scale Chinese studies show that substantial gains can still be unlocked through technology upgrades, pollutant control, waste classification, and load-rate improvements, but they do so within already-generating systems rather than at the point of first entry (Liu et al., 2025; Han et al., 2025). The present results are consistent with persistent generator hierarchy rather than natural convergence after entry.

Table 3: Core electricity-recovery specifications in the canonical generator frame

Variable	Model 1 Pooled OLS	Model 2 Year FE	Model 3 RE	Model 4 Year FE + RE
Facility age (years)	-0.0279***	-0.0348***	-0.0188***	-0.0332***
	(0.0022)	(0.0022)	(0.0025)	(0.0021)
Capacity (100 t/day)	0.0874***	0.1030***	0.0405***	0.0519***
	(0.0083)	(0.0086)	(0.0083)	(0.0096)
Capacity utilization	0.7468***	0.7789***	0.6199***	0.5411***
	(0.1421)	(0.1346)	(0.0997)	(0.0943)
Heating value (MJ/kg)	0.0010	0.0032	0.0006	0.0012
	(0.0023)	(0.0021)	(0.0012)	(0.0010)
Grid EF (kg-CO_2/kWh)	0.3182	-0.4466	1.6333***	-0.1951
	(0.2219)	(0.2714)	(0.1965)	(0.2101)
Observations	5,683	5,683	5,683	5,683
Facilities	1,016	1,016	1,016	1,016
R-squared	0.2470	0.3721	0.1647	0.3076

Note: OLS means ordinary least squares, FE means fixed effects, RE means random effects, EF means emissions factor, MJ/kg means megajoules per kilogram, and kg-CO_2/kWh means kilograms of carbon dioxide per kilowatt-hour. Standard errors are clustered by facility and reported in parentheses. Three asterisks denote $p < 0.01$. Coefficients are reported as structured conditional associations rather than as strict structural parameters. Because the dependent variable is logged, small coefficients can be read approximately as percentage changes in electricity recovered per tonne.

This is why the intensive margin cannot be inferred from the adoption margin alone. A facility can be inside generation without being close to the frontier. At the same time, operations alone are unlikely to erase large vintage and scale gaps once plants are mature. Conditional performance is therefore a structured generator-performance problem rather than a simple continuation of the entry problem.

For interpretation, this matters as much as the adoption result. If the paper only documented selective entry, a reader could infer that the main task was simply to push more facilities into generation. If it only documented generator hierarchy, a reader could infer that non-generators were just lagging versions of the same problem. The linked result rejects both shortcuts. The fleet appears divided between a segment that still struggles to enter generation and a segment where entry has occurred but performance remains uneven and structured.

4.3 Why the two results belong together

Read together, the two margins change the modernization story. Some facilities still need to enter energy recovery, while others already generate but remain far apart in performance. The adoption results show selective entry before generator performance is considered. The electricity-recovery results show that large gaps inside the generating segment are not easily erased through within-facility movement alone. A one-average-fleet model would flatten those margins into a single modernization narrative and understate both the selectivity of entry and the persistence of cross-facility performance differences. The point is not that the two samples form one strict causal chain. It is that they identify different constraints within the same fleet: who gets into generation, and how far operating generators can move once they are already there.

5 Discussion

The paper's main interpretive claim is methodological: in this fleet, entry into generation and performance after entry are different problems. Modeling them separately prevents the non-generating segment and the mature generating segment from being collapsed into one muted fleet mean. That separation reveals the combination of selective entry and persistent hierarchy that structures the system.

The substantive interpretation is correspondingly two-part. On the adoption margin, the data do not support broad late conversion among old small plants. Observed transition within the coded at-risk frame is concentrated among younger and larger facilities, and the pathway audit is more consistent with a capital-intensive event mix than with diffuse late-life catch-up. On the electricity-recovery margin, age, scale, and utilization still matter strongly within the canonical regression frame, while within-facility movement remains modest relative to the cross-sectional hierarchy. Read together, the evidence points more toward selective entry into the generating regime and persistent generator hierarchy than toward easy convergence once entry has occurred.

This makes the paper useful as a planning diagnostic rather than as an intervention ranking. Table 4 translates the empirical patterns into defensible planning questions while keeping the claim boundaries explicit.

Table 4: Planning diagnostic implied by the two-margin evidence

Empirical pattern	Planning use	Boundary condition
Younger/larger entry	Screen non-generators by age, scale, and replacement timing before assuming a simple retrofit path	Older or smaller plants are not proven unable to upgrade
Mixed event pathways	Check asset histories before treating adoption as one retrofit mechanism	The audit does not distinguish replacement, refurbishment, and reporting change with certainty
Structured generator performance	Compare generators with similar age, scale, and utilization before judging improvement room	Coefficients are conditional associations, not causal effects
Low within-facility movement	Separate incremental operating improvements from capital-renewal or consolidation decisions	Plant-specific operational gains remain possible

The interpretation has clear limits. The pathway audit does not prove that replacement is the unique pathway of modernization, and the regressions do not provide strict causal estimates of vintage lock-in or clean estimates for all operating generators in Japan. Alternative interpretations remain possible, including reporting compression, unobserved retrofit histories, unmeasured governance differences, and institutional constraints that limit operational responses. The defended claim is therefore narrower: the data support a selective observed-entry pattern and persistent generator performance hierarchy, not a uniquely identified mechanism or a full causal hierarchy. The supplement is used to make those limits auditable: it records the pathway rules, identifier sensitivity, heating-value restrictions, and estimator variants rather than leaving those judgments implicit.

The results do not identify the best intervention for any individual municipality. They indicate that planning assessments should first distinguish facilities outside electricity recovery from operating generators, because the observable constraints differ across those two groups. For the non-generating energy-recovery segment, especially older non-generators and small plants, the evidence supports diagnostic screening for whether renewal, consolidation, or continued non-generation is plausible. For the already-generating segment, the evidence supports a different diagnostic question: whether an existing generator has realistic room for operational or capital improvement within its observed age, scale, and utilization context.

That distinction is administrative as well as technical. Japanese waste systems are organized through municipalities whose planning boundaries do not always align with efficient waste sheds, and intermunicipal reorganization has its own political costs and institutional legacies (Rausch, 2006; Sakai et al., 2008). In that setting, collapsing non-generators and generators into one average segment risks hiding the difference between entry-side asset decisions, which are lumpy and governance-heavy, and generator performance assessment, which is more incremental and operational.

The broader policy context points in the same direction. Comparative waste policy studies and European framework discussions both treat waste-to-energy as valuable only when embedded inside a wider hierarchy that preserves prevention, reuse, and recycling priorities (Sakai et al., 2011; European Commission, 2017). The empirical split helps explain why those hierarchies matter. A municipal system can have real energy-recovery gains available at the non-generating margin while still facing a different, narrower set of decisions inside the already-generating segment.

For climate interpretation, the relevant question is not simply whether a plant generates, but what kind of generator it is and how much improvement remains inside that segment. Waste-to-energy performance is judged against both an internal engineering standard and the emissions profile of the broader energy system, including the avoided-emissions logic built into carbon accounting (Astrup et al., 2009; Münster & Meibom, 2010). In Japan's current decarbonization setting, where national inventories and scenario work track sectoral emissions and energy mix changes, the two-part design is useful because it keeps those questions separate (Greenhouse Gas Inventory Office of Japan & Ministry of the Environment Japan, 2024; Yamada et al., 2023).

That separation is also what makes the paper more understandable for a planning audience. Municipal systems rarely choose among abstract technological ideals. They decide whether to renew an aging non-generator, coordinate waste routing toward a larger plant, maintain an existing generator, or invest in an upgrade for a plant that already produces electricity. Those are related decisions, but they are not interchangeable. Keeping the extensive and intensive margins separate helps planners locate the bottleneck without relying on one blended fleet mean.

6 Conclusion

In the observed facility panel, Japan's electricity-recovery modernization does not appear as one smooth fleet-wide process. Within the coded adoption frame, observed entry into generation is selective rather than diffuse. Within the canonical generator frame, electricity recovery performance remains stratified by age, scale, and utilization, with limited within-facility movement relative to between-facility differences. Read together, those margins show why an aggregate fleet view can misstate the modernization bottleneck. The paper does not identify one unique pathway or intervention hierarchy, but it does show why municipal fleet studies gain by separating adoption from conditional performance. For planners, the first question is whether renewal, consolidation, or continued non-generation is justified for a facility outside electricity recovery; the second is whether an existing generator has realistic room for operational or capital improvement.

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Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval

This study uses publicly released administrative facility data and does not involve human participants, animal subjects, or private personal data.

Data Availability

The facility-level source data are derived from the Ministry of the Environment Japan General Waste Treatment Survey, which is publicly released by the Ministry of the Environment. The cleaned analysis outputs, figure-generation scripts, manuscript figures, and reproducible paper workspace can be made available by the author on reasonable request, subject to any redistribution limits attached to the source administrative files.

Generative Artificial Intelligence (AI) and AI-Assisted Technologies Statement

During the preparation of this manuscript, the author used OpenAI Codex and Anthropic Claude to support drafting, language revision, and organizational planning. These tools were not used as authors, did not generate or verify the underlying data, and did not replace author review. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

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